

Pre-class preparation

Please read the following textbook sections from Degroot and Schervish's *Probability and Statistics* (fourth edition) or watch the video, as indicated:

- Textbook: Section 8.4 (can skip part about Cauchy). Also read the part at the end with the derivation of the t_{df} PDF!
 - Be sure you understand the all steps of derivations of the distributions in this section! If you don't, come to office hours to ask!

Objectives

By the end of the day's class, students should be able to do the following:

- Derive and give the PDF of the t_{df} distribution.
- Understand how the t distribution compares to the standard Normal distribution.
- Understand when we need the t distribution when conducting inference on the mean for a random sample from $N(\mu, \sigma^2)$.

Reflection Questions

Please submit your answers to the following questions to the corresponding Canvas assignment by 9:00AM:

1. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$ with both parameters unknown. Let $\bar{X}$ be the sample mean, $S = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}$. What are the distributions of $\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma}$ and $\frac{\sqrt{n}(\bar{X} - \mu)}{S}$?
2. The measurement errors of a particular device are independently Normally distributed with mean μ and variance σ^2 . (We would hope $\mu = 0$, but we don't actually know!) We use the device five times, and each measurement i 's error is X_i . Find an expression for the probability that the sample mean will be within $\frac{2}{\sqrt{n}}S$ units from the true mean, and then approximate this probability using R.

Some useful R code: If $Z \sim N(0, 1)$, then $\Phi(z) = P(Z \leq z) = \text{pnorm}(z)$. If $T \sim t_{df}$, then $P(T \leq t) = \text{pt}(t, df)$.

3. (Optional) Is there anything from the pre-class preparation that you have questions about? What topics would you like would you like some more clarification on?